Anesthetic management of thoracic trauma

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Purpose of review

Trauma remains a leading cause of death across all age groups. Thoracic injury is a contributing cause in approximately half of these. Despite being potentially life threatening, most thoracic trauma is managed nonoperatively or with an intercostal catheter. Only 10% of thoracic trauma patients will require emergency thoracotomy. Many more will undergo emergency or urgent surgical intervention for coexisting injuries. Thoracic injuries are dynamic. It is crucial for the anesthesiologist to continually reassess the patient, so that the manifestations of evolving injuries may be detected as early as possible and appropriate management decisions made. Up-to-date knowledge of injury patterns, mechanisms, pathophysiology, and operative and nonoperative management will facilitate optimal management of these patients.

Recent findings

There is recent literature discussing the surgical, anesthetic and critical care management of a range of thoracic injuries resulting from either blunt or penetrating trauma.

Summary

Initial resuscitation and surgical management of patients with thoracic trauma continue to evolve. Improvements in prehospital care and diagnostic techniques as well as development of minimally invasive interventions mean that the anesthesiologist may be required to provide care to unstable patients in an expanded range of scenarios and environments.

Keywords

aortic injury, blunt cardiac injury, thoracic injuries, tracheobronchial injury

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Introduction

Thoracic trauma accounts for 25–50% of all trauma deaths in the US [1••]. Mortality from chest injuries is around 10% overall [2]. The semi-rigid thorax protects vital organs including the heart, lungs, major vessels and airways. Significant force is usually required to injure these structures.

While the majority of patients with thoracic trauma can be managed conservatively or with a simple intercostal catheter (ICC) [3•], a small but significant number of blunt (10%) and penetrating (15–30%) injuries [2,3•] require thoracotomy as a component of initial resuscitation. Often patients will have multiple coexisting injuries also requiring urgent surgery.

Airway injury

Intrathoracic tracheobronchial injuries are less common than upper airway injuries. They are present in 0.8% of patients presenting for emergency surgery after blunt thoracic trauma [4]; 76% occur within 2.5 cm of the carina [5], a relatively fixed point. Their high mortality is related to ventilation/oxygenation difficulties [6••]. The diagnosis is often delayed [5].

Intrathoracic airway injuries should be managed in consultation with a thoracic surgeon as operative repair is usually required [6••], although good results have recently been reported with conservative management of stable injuries after careful assessment [7•]. The goal is to improve ventilation and reduce the effects of air leak into surrounding structures by placing a cuffed airway distal to the injury. A definitive airway can usually be secured using a fiber-optic bronchoscope, while maintaining spontaneous ventilation. Rigid bronchoscopy may be required if there is severe airway disruption and bleeding, but this is contraindicated in patients with an unstable cervical spine.

Blunt cardiac injury

The reported incidence of blunt cardiac injury following chest trauma is between 5 and 50%, reflecting the variation in diagnostic criteria [8,9]. The spectrum of cardiac injury varies from isolated ECG changes to cardiac rupture. The right ventricle and interventricular

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septum are most frequently involved [9]. The main concerns are the risk of arrhythmias or ventricular dysfunction which may occur as a result of impaired contractility or structural damage to the valves or chamber walls.

Arrhythmias on presentation (commonly ventricular ectopics) or ST-segment changes should arouse suspicion of a cardiac injury. In a patient with otherwise minor trauma, a normal ECG at admission can exclude the risk of significant blunt cardiac injury, eliminating the need for cardiac monitoring if not otherwise indicated [10]. If the admission ECG is abnormal the patient should have continuous ECG monitoring for 24–48 h. If the patient is hemodynamically unstable, an echocardiographic assessment of preload, myocardial contractility, pericardial collections, and myocardial and valvular structures should be performed. Other imaging including coronary angiography may occasionally be undertaken, although nuclear medicine scans probably offer little useful further information [10]. Severe arrhythmias may develop secondary to injury to areas of myocardium too small to lead to increased levels of markers for myocyte damage. Measurement of troponin probably has little value in the management of blunt cardiac injury [11]. It may be prudent to delay nonurgent surgery for a period of 24–48 h after significant injury to allow some recovery of the abnormal myocardium and conducting system. Intra-aortic balloon counter-pulsation should be considered if there is cardiogenic shock [8].

Penetrating cardiac injury

Penetrating anterior chest wounds causing cardiac injury are typically fatal, with only 6% of patients surviving to reach hospital [12•]. In-hospital mortality is significantly higher after gunshot wound to the heart (81.5%) than after stab wound (19.8%). Presentation with tamponade is common after stab wound [12•]. Beck's classical triad of distended neck veins, hypotension and quiet heart sounds may not be present if there is severe hypovolemia. Transesophageal echocardiography (TEE) assists rapid diagnosis in most cases. Once the diagnosis is made, emergency left anterior thoracotomy is preferable to subxiphoid needle pericardiocentesis if the surgical skills are available. Rapid fluid preloading and cautious induction of anesthesia is appropriate, although *in extremis* the goal may be amnesia (e.g. titrated doses of midazolam and fentanyl). Those patients with vital signs absent for less than 5–10 min are candidates for emergency resuscitative thoracotomy (see below).

Thoracic injury and video-assisted thoracoscopic surgery

Thoracoscopic techniques have an expanding role in both the acute and subacute management of acute chest injury, and offer the potential for reduced postoperative pain and enhanced recovery compared with open surgery [13••].

Video-assisted thoracoscopic surgery (VATS) is a useful technique for dealing with late complications such as retained hemothorax, posttraumatic empyema and bronchopleural fistula. VATS requires lung isolation, which can be achieved with a double-lumen endobronchial tube, a bronchial blocker or a Univent tube (Fuji Systems, Tokyo, Japan). VATS remains unsuitable and contraindicated in cases of significant hemodynamic instability, or with inability to tolerate single-lung ventilation or lateral decubitus position. Furthermore, VATS should not be undertaken when there is a major injury to the heart or great vessels and is impossible if the pleural space is obliterated.

Hemothorax

Sources of bleeding in the pleural cavity depend upon the mechanism of injury, but include lung laceration, intercostal or internal thoracic artery injury and great vessel tears. It may be difficult to clinically assess the volume of a hemothorax. An ICC should be sited early – certainly well before signs of shock become apparent, by which time up to half the blood volume may have been lost into one hemithorax [14••]. Rapid drainage of greater than 1500 ml blood from the ICC after insertion (massive hemothorax) is an indication for urgent thoracotomy [15]. In patients who are relatively stable, with less severe ongoing bleeding (e.g. less than 200 ml/h from the ICC for 2–4 h or less than 1500 ml in 24 h), VATS can control ongoing hemorrhage with a success rate of up to 82% [13••].

Pneumothorax

A persistent pneumothorax occurs when there is ongoing air leak and failure to fully re-expand the lung with 72 h of suction via an ICC. This may occur in up to 23% of trauma patients [16]. Such leaks may persist over a period of months. VATS can be used to identify and resect the sites of air leak. Small pneumothoraces, such as those detected on computed tomography scan, have been managed conservatively, without ICC [17]. It is important to be aware that these pneumothoraces may rapidly develop tension with positive pressure ventilation.

Diaphragmatic injury

Studies have shown that VATS has a sensitivity and specificity for detecting diaphragmatic injury of 97–100%, with an accuracy of about 98% [13••]. This is significantly higher than combinations of physical examination, chest radiograph, computed tomography, focused abdominal sonogram in trauma (FAST) and diagnostic peritoneal lavage.

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Aortic injuries

Blunt aortic injury, typically caused by high-speed motor vehicle accidents, is the second most common cause of death in blunt trauma victims after severe head injury [18]. It is nearly always associated with other injuries. A classic study of 275 cases of aortic rupture published in 1958 reported that 85% die at the scene [19]. Half of those that survive to hospital die within 24 h [20,21•]. The most common site of injury is at the aortic isthmus just distal to the origin of the left subclavian artery because of tethering of the ligamentum [22]. Survivors have intact adventitial layers or contained hematoma within the mediastinum and pseudoaneurysm, although blunt aortic injury can be limited to subtle intimal flaps, which are challenging to detect radiologically unless there is coexisting dissection [23•].

A high index of suspicion must be maintained for timely diagnosis of blunt aortic injury. Although chest radiograph remains a useful initial investigation, helical or multidetector computed tomography imaging is replacing aortography as the definitive screening test for evaluating aortic tears because it is highly accurate [23•]. TEE may also be useful, but the trachea and left main bronchus interferes with aortic arch assessment.

Low mortality rates, excellent technical success and shorter hospital stay mean that endovascular stent grafting (EVSG) may be evolving into first-line treatment for aortic injury [22,24•,25•,26]. Conventional immediate open surgery and repair is associated with mortality rates of up to 28% [27] with paraplegia in a further 6–19% depending on the surgical technique [28]. There have been advances over the past decade, and advocates of both passive and active methods of distal aortic perfusion during cross-clamping have reported decreased spinal cord ischemia and reduced mortality rates [24•,29•]. Although there are no randomized trials comparing EVSG with open surgical repair of blunt aortic injury, nearly all reported series underscore significant early and mid-term advantages of endovascular treatment, at least in selected cases. The timing of treatment remains controversial [21•,24•,25•,30••], but if the patient has more immediately life-threatening injuries then EVSG can be delayed [22]. A recent review of traumatic aortic rupture utilizing EVSG identified 284 patients in total [30••]. The overall rate of procedure-related complications was 12% (vs. 20% with open surgery) including several strokes, but there was no paraplegia. Paraplegia is certainly described after elective descending thoracic endograft [31•]. Overall mortality was 4% for EVSG vs. 18.9% for open surgical repair, although the authors acknowledged the possibility of publication bias. Overall survival without repair is thought to be only 2% at 3 months [32], although conservative management utilizing blood pressure control may be the best option for mild injuries such as isolated intimal flaps [27]. It has also been described in patients with severe coexisting injuries whose aortic injury is stable [33••]. It should be noted that long-term results are not yet available for EVSG. Although EVSG is performed in the radiology suite, the anesthetic principles and standards of monitoring are the same as those generally applicable to open thoracic aortic surgery.

The anesthetic goals for open aortic surgery are to reduce shear stress in the aortic walls by meticulous blood pressure control (systolic blood pressure less than 140 mmHg) using β-blockade and glyceryl trinitrate. Aggressive treatment with sodium nitroprusside during proximal aortic cross-clamping may be associated with a higher incidence of postoperative paraplegia [34]. Duration of cross-clamping is a key determinant of outcome and spinal cord protection is critical. Mild hypothermia (e.g. 34°C) during cross-clamping may provide some benefit, but rewarming should be undertaken as soon as possible to avoid complications of hypothermia such as coagulopathy. Inability to tolerate single lung ventilation, systemic heparinization or increased intracranial pressure due to aortic cross-clamping may all complicate operative management.

In penetrating aortic trauma, permissive hypotension (e.g. systolic blood pressure 90 mmHg) until definitive surgical control may avoid further hemorrhage, coagulopathy, acidosis and hypothermia [1••].

Pulmonary contusion–flail chest complex

Pulmonary contusion, present in 25–35% of blunt chest trauma, is often associated with flail chest [35], and is a risk factor for acute respiratory distress syndrome, pneumonia and long-term respiratory dysfunction. Although mortality of 10–25% has been reported [36•], these deaths may be related to severity of the trauma and associated injuries. Cohn [37] described the clinico-pathological correlate of a pulmonary contusion. Parenchymal hemorrhage, interstitial edema and decreased surfactant production result in alveolar collapse and consolidation. These contribute to ventilation/perfusion mismatch, pulmonary shunting and decreased lung compliance. Clinical signs include tachypnea, hypoxemia, hypercarbia, wheezing and hemoptysis. Chest radiograph changes are often delayed. The contusion may appear two to three times larger on computed tomography scan. Soldati *et al.* [36•] describe the use of ultrasound in the diagnosis of pulmonary contusions with positive and negative predictive values (compared with computed tomography as gold standard) of 94 and 96%.

The primary goal in the management of pulmonary contusions is the maintenance of adequate oxygenation.

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Treatment modalities include noninvasive and invasive ventilation, high-frequency and differential lung ventilation, extracorporeal membrane oxygenation (ECMO) [38•], and surfactant. Mechanical ventilation utilizing low (6 ml/kg) tidal volumes, high levels of positive end-expiratory pressure and permissive hypercarbia [39] may protect the lung from further insult. Optimal pain control is an important goal in the management of significant chest wall injuries, to enable effective deep breathing and coughing. Epidural analgesia may be useful. Large flail segments may require mechanical ventilation, the duration of which is possibly shortened by operative fixation of the rib fractures [40•]. Fracture of the first and second ribs requires major force, and should always prompt a search for significant intrathoracic trauma including aortic injury.

Sudden cardiovascular or neurological dysfunction, dyspnea and hemoptysis occurring after initiation of positive pressure ventilation in thoracic trauma victims is a classic presentation of systemic air embolism which may occur if there is a bronchial–pulmonary communication. TEE may be diagnostic. Management includes minimizing inspiratory pressures, intravascular volume loading, dependent positioning of the injured lung and thoracotomy with hilar clamping.

Emergency resuscitative thoracotomy is a temporizing measure of most benefit to victims of penetrating chest trauma who arrive with, at least, signs of life such as agonal ventilation, pupillary reactivity or cardiac electrical activity [2]. The overall survival after emergency resuscitative thoracotomy is 7%. The outcome in blunt trauma victims is dismal, whereas in stab victims with isolated cardiac injury survival may be as good as 20–30% [3•]. The left anterior approach enables evacuation of pericardial tamponade, direct control of hemorrhage from the heart, lung or great vessels, open cardiac massage, cross-clamping of the aorta to arrest massive blood loss from below the level of the diaphragm, and left hilar clamping for massive systemic air embolism and bronchopleural fistula.

Pediatric chest trauma

In children, significant thoracic injuries can occur with few external signs as a result of their highly compliant chest wall. Although the general principles for management of pediatric chest trauma are similar, there are key anatomical, physiological and pharmacological differences which influence treatment [41].

Anesthetic management

Despite different patterns of injury with blunt and penetrating trauma, cardiorespiratory instability, tissue hypoxia and acidosis are commonly established on presentation. Airway obstruction, ventilatory insufficiency including tension pneumothorax or cardiovascular compromise due to hypovolemia, pericardial tamponade or cardiogenic shock will rapidly cause the patient's demise without timely and appropriate management. Frequent reassessment is important, so that manifestations of evolving injuries may be detected as early as possible and appropriate management decisions made. The anesthesiologist should be present when the patient arrives in the emergency room, where initial management follows Advanced Trauma Life Support principles. A primary survey is undertaken while the patient's vital signs are rapidly assessed, a history obtained and initial trauma investigations begun.

Preoxygenation with 100% oxygen for 3–4 min is not always possible in the agitated trauma patient. The standard approach for securing a definitive airway is direct laryngoscopy and rapid sequence intubation with cricoid pressure and manual in-line immobilization of the cervical spine after clinical assessment of the airway. Unanticipated difficult intubation should be managed in accordance with the American Society of Anesthesiologists Difficult Airway Algorithm (Trauma Version) [42]. Worsening hypotension and difficulty with ventilation should prompt consideration of a diagnosis of tension pneumothorax until proven otherwise.

The general principles of trauma anesthesia such as oxygenation, restoration of circulating volume, correction of hypothermia and coagulopathy certainly apply to patients with thoracic injuries. Many institutions do not undertake lung isolation techniques for emergency trauma thoracotomy. Double-lumen tubes are often difficult to insert during rapid sequence intubation and are commonly associated with malposition. Anesthesiologists with limited experience in thoracic anesthesia fail to successfully place lung isolation devices in over a third of patients [43••]. The failure rate is independent of the type of lung isolation device. A double-lumen endobronchial tube should be changed to a conventional endotracheal tube should postoperative ventilation be required. This can be avoided by utilizing a Univent tube or a bronchial blocker system [44••]. Fiber-optic bronchoscopy is required to optimally position a bronchial blocker and significant bleeding in the tracheobronchial tree may preclude its use.

The choice of drug for induction of anesthesia is less important than using a dose appropriate for the cardiovascular status of the patient. Etomidate or thiopental usually represent a better choice than propofol [45]. Ketamine is not widely used because its direct myocardial depressant effect may predominate in the unstable trauma patient with maximal sympathetic activity. Hypotension postinduction should be promptly

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treated with sympathomimetic drugs and ongoing volume resuscitation. Maintenance using a low-dose volatile agent with fentanyl and nondepolarizing muscle relaxant is usually suitable. Nitrous oxide should be avoided owing to increases in the volume of gas-containing lesions such as pneumothoraces and air emboli.

Monitoring should include direct arterial and central venous pressures and, if indicated, TEE. In addition, an awareness monitor such as the bispectral index monitor is extremely useful in the setting of trauma as this can help to avoid both under- and overdosing of anesthetic drug administration.

Conclusion

Blunt or penetrating thoracic injuries present a particular challenge as a result of complex, dynamic and life-threatening injuries. Despite this, these often young patients frequently make a full recovery. The role of the anesthesiologist, a key member of a multidisciplinary trauma team, is critical.

References and recommended reading

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Additional references related to this topic can also be found in the Current World Literature section in this issue (pp. 87–88).

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